

Lab 2: Vector Arithmetic, Lengths, and the Dot Product

Math 124, Fall 2026 at the University of Michigan

due by the end of class on Wednesday, September 9, 2026

Name: _____

username: _____

Each lab worksheet will contain several activities, most of which will involve writing math on paper, and some of which will involve running code in a Jupyter Notebook. Lab activities are meant to last an hour, and the second hour of lab is dedicated to starting the homework assignment. To receive credit for a lab, you must show your lab TA your work on both the lab worksheet and homework assignment.

While you must get checked off by your lab TA **individually**, we encourage you to form groups with 1-2 other students to complete the activities together.

Review

- $\vec{v} \in \mathbb{R}^n$, pronounced “v in R n”, means that $\vec{v}$ is a vector with n components. Usually, $v_1, v_2, \dots, v_n$ are placeholders for $\vec{v}$ ’s components.
- A vector that results from scaling and adding one or more vectors is called a **linear combination** of the original vectors. For example, if $\vec{u} = \begin{bmatrix} 5 \\ 1 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 1 \\ 7 \end{bmatrix}$, then $2\vec{u} - 3\vec{v}$ is a linear combination of $\vec{u}$ and $\vec{v}$. It is the new vector

$$2\vec{u} - 3\vec{v} = 2 \begin{bmatrix} 5 \\ 1 \end{bmatrix} - 3 \begin{bmatrix} 1 \\ 7 \end{bmatrix} = \begin{bmatrix} 10 - 3 \\ 2 - 21 \end{bmatrix} = \begin{bmatrix} 7 \\ -19 \end{bmatrix}.$$

- For a vector $\vec{v} \in \mathbb{R}^n$, its **length**, **norm**, or **magnitude**, is

$$\|\vec{v}\| = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}.$$

For the vectors above,

$$\|\vec{u}\| = \sqrt{5^2 + 1^2} = \sqrt{26}, \quad \|\vec{v}\| = \sqrt{1^2 + 7^2} = \sqrt{50}.$$

- The **dot product** of two nonzero vectors $\vec{u}, \vec{v} \in \mathbb{R}^n$ can be computed in two ways:
 - Algebraic definition: $\vec{u} \cdot \vec{v} = u_1v_1 + u_2v_2 + \dots + u_nv_n$.
 - Geometric definition: $\vec{u} \cdot \vec{v} = \|\vec{u}\|\|\vec{v}\|\cos\theta$, where θ is the angle between the vectors.

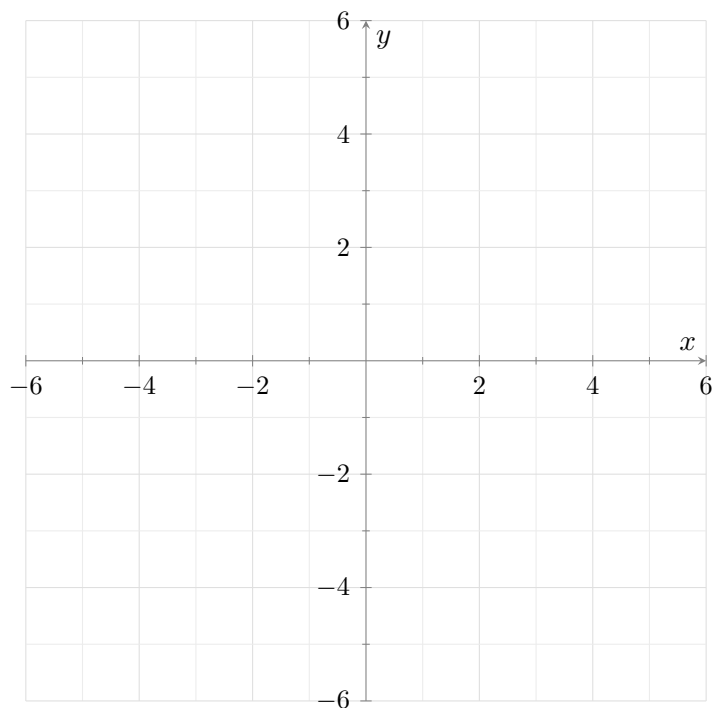
For the vectors above,

$$\vec{u} \cdot \vec{v} = \begin{bmatrix} 5 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 7 \end{bmatrix} = 5(1) + 1(7) = 12.$$

Activity 1: Linear Combinations and Lengths

Let $\vec{u} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} -1 \\ -4 \end{bmatrix}$.

- a) Let $\vec{w} = -\vec{u} - \vec{v}$ and let $\vec{x} = 2\vec{u} + \vec{v}$. Draw $\vec{u}$, $\vec{v}$, $\vec{w}$, and $\vec{x}$ on the axes below, with each vector starting at the origin. Label each vector.

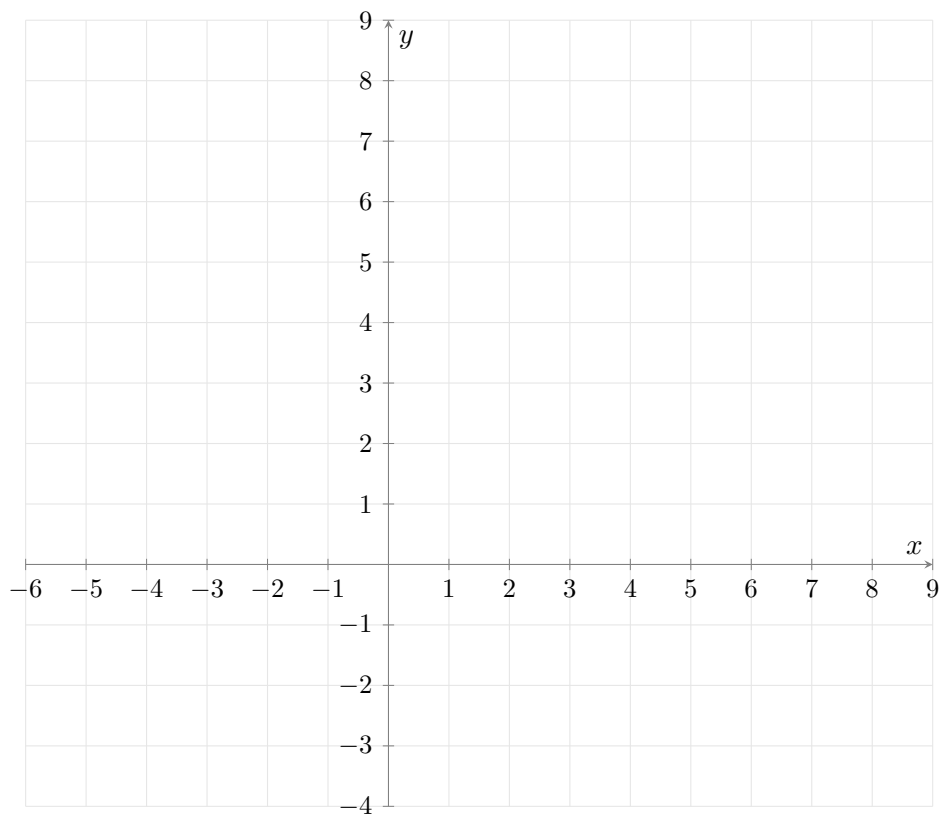


- b) Which of the following vectors is equal to $5\vec{e}_1 - 4\vec{e}_2$? Select one option. Recall that $\vec{e}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\vec{e}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$ are the **standard basis vectors** in $\mathbb{R}^2$.
- ☐ $3\vec{u} - 4\vec{v}$
 ☐ $4\vec{u} + 3\vec{v}$
 ☐ $3\vec{u} + 4\vec{v}$
 ☐ $-3\vec{u} + 4\vec{v}$
 ☐ $4\vec{u} - 3\vec{v}$

- c) Find the lengths of $\vec{u}$ and $\vec{v}$.

Activity 2: The Dot Product

For each pair of vectors below, draw them on the axes, compute their dot product, and select whether the angle between them is acute, right, or obtuse. Draw each vector from the origin.



a) $\begin{bmatrix} 8 \\ 6 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

☐ Acute ☐ Right ☐ Obtuse

b) $\begin{bmatrix} 8 \\ 6 \end{bmatrix}$ and $\begin{bmatrix} -5 \\ 0 \end{bmatrix}$

☐ Acute ☐ Right ☐ Obtuse

c) $\begin{bmatrix} 8 \\ 6 \end{bmatrix}$ and $\begin{bmatrix} 6 \\ 8 \end{bmatrix}$

☐ Acute ☐ Right ☐ Obtuse

d) $\begin{bmatrix} 8 \\ 6 \end{bmatrix}$ and $\begin{bmatrix} 4 \\ 7 \end{bmatrix}$

☐ Acute ☐ Right ☐ Obtuse

e) $\begin{bmatrix} 8 \\ 6 \end{bmatrix}$ and $\begin{bmatrix} -3 \\ 4 \end{bmatrix}$

☐ Acute ☐ Right ☐ Obtuse

Activity 3: Angles and Orthogonality

Suppose $\vec{u} = \begin{bmatrix} 5 \\ 0 \\ -4 \\ 1 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 9 \\ 1 \\ 2 \\ 3 \end{bmatrix}$.

- a) Find $\vec{u} \cdot \vec{v}$, $\|\vec{u}\|$, and $\|\vec{v}\|$.

- b) Using the results of part a), find the angle between $\vec{u}$ and $\vec{v}$. Leave your answer in the form $\cos^{-1}(\cdot)$.

Even though we cannot directly visualize these vectors in $\mathbb{R}^4$, the formula

$$\theta = \cos^{-1}\left(\frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\|\|\vec{v}\|}\right)$$

defines the angle between them. Another term for the cosine of the angle between two nonzero vectors is their **cosine similarity**.

Activity 4: Comparing Vectors

As we saw in the previous activity, the cosine similarity of two nonzero vectors is the cosine of the angle between them. It is one of the many ways we can measure how “different” two vectors are, by comparing their directions.

Suppose

$$\vec{u} = \begin{bmatrix} 2 \\ 1 \\ 2 \end{bmatrix} \quad \text{and} \quad \vec{v} = \begin{bmatrix} -2 \\ 4 \\ 4 \end{bmatrix}.$$

Explore and rotate the vectors at desmos.com/3d/4wygwwruil.

- a) Find the cosine similarity of $\vec{u}$ and $\vec{v}$.

- b) Another way to measure how “different” two vectors are is to measure the distance between their tips when both vectors start at the origin. Find this distance for $\vec{u}$ and $\vec{v}$, that is, $\|\vec{u} - \vec{v}\|$.